Project 1: Power BI and Data Set - Project Report

1. Introduction

This report documents the approach, methodology, and findings for Project 1: Power BI and Data Set. The project was undertaken using Microsoft Power BI Desktop and involved end-to-end data preparation, metric computation, and dashboard visualisation of an insurance agency dataset.

2. Objective

The objective of this project is to improve the agents' existing knowledge used for agent segmentation in a supervised predictive framework. The analysis focuses on evaluating agent performance across multiple dimensions including premium growth, policy retention, loss ratios, and quoting system efficiency.

3. Problem Statement

An insurance group consisting of 10 property and casualty insurance, life insurance, and insurance brokerage companies operates across a multi-state region. The group is represented by more than 4,000 independent agents operating throughout a six-state region. The motivation for this analysis is to evaluate agent performance based on attributes such as demography, products sold, new business written, and retention of existing policies, with the ultimate goal of improving agent segmentation models.

4. Dataset Overview

The dataset used was **finalapi.csv**, containing insurance agency performance data from 2005 to 2015. The original dataset contained **44 columns** and over **213,000 rows.**

4.1 Data Cleaning Steps

The following cleaning steps were performed in Power Query using M code:

Step	Action
Column Selection	Removed irrelevant columns, retaining 29 columns aligned to brief requirements
Extra CSV Columns	Identified and removed 5 additional columns generated by CSV formatting inconsistencies
Year Filter	Filtered data to include only years 2006 to 2013 as required by the brief
Data Type Conversion	Set correct data types for all columns — text, whole number, and decimal using en-US locale to resolve regional formatting issues

Placeholder Replacement	Replaced all 99999 placeholder values with null across all columns
LOSS_RATIO Cleaning	Replaced values above 10 and below 0 with null to remove invalid entries

4.2 Key Data Quality Findings

Column	Issue Found	Resolution
RETENTION_RATIO	60.93% placeholder values (99999)	Replaced with null
LOSS_RATIO	24.02% placeholders + negative values	Replaced with null
PREV_WRTN_PREM_AMT	6.51% placeholders	Replaced with null
CL columns	46.26% placeholders	Replaced with null — agencies not writing CL business
PL columns	26.93% placeholders	Replaced with null — agencies not writing PL business

5. Computed Metrics

The following metrics were computed and added as independent attributes to the dataset as required by the brief:

5.1 Growth Rates (2006–2013)

Two growth rate columns were computed and added directly to the data model in Power Query:

Premium Growth Rate:

Premium_Growth_Rate = (WRTN_PREM_AMT - PREV_WRTN_PREM_AMT) / PREV_WRTN_PREM_AMT

Policy Growth Rate:

Policy_Growth_Rate = (POLY_INFORCE_QTY - PREV_POLY_INFORCE_QTY) / PREV_POLY_INFORCE_QTY

Both columns handle division by zero and null values, returning null where the previous year value is unavailable.

5.2 Hit Ratios by Quoting System

Hit ratios were computed as DAX measures for each of the 8 quoting systems:

Measure	**Formula**
Hit_Ratio_CL_MDS	Bound / Quoted (CL-MDS)
Hit_Ratio_CL_SBZ	Bound / Quoted (CL-SBZ)
Hit_Ratio_CL_eQT	Bound / Quoted (CL-eQT)
Hit_Ratio_PL_ELINKS	Bound / Quoted (PL-ELINKS)
Hit_Ratio_PL_PLRANK	Bound / Quoted (PL-PLRANK)
Hit_Ratio_PL_eQTte	Bound / Quoted (PL-eQTte)
Hit_Ratio_PL_APPLIED	Bound / Quoted (PL-APPLIED)
Hit_Ratio_PL_TRANSACTNOW	Bound / Quoted (PL-TRANSACTNOW)

5.3 Product Line Hit Ratios

Two additional measures were created to aggregate hit ratios at product line level:

- **Hit_Ratio_CL** — Combined hit ratio across all 3 Commercial Lines systems
- **Hit_Ratio_PL** — Combined hit ratio across all 5 Personal Lines systems

6. Visualisations

The following visualisations were created on the Power BI dashboard:

Page 1 — Core Analysis

Visual Type Fields Used

1 Summary Statistics by Agency Matrix Table AGENCY_ID, WRTN_PREM_AMT, RETENTION_RATIO, ACTIVE_PRODUCERS, NB_WRTN_PREM_AMT, LOSS_RATIO

2 Premium by Product Line Clustered Column Chart PROD_LINE, Total_Written_Premium

3 New Business by Product Line Pie Chart PROD_LINE, NB_WRTN_PREM_AMT

4 Top 10 Agencies by Retention Ratio Horizontal Bar Chart AGENCY_ID, Avg_Retention_Ratio

5 Retention Ratio by State Horizontal Bar Chart STATE_ABBR, Avg_Retention_Ratio

6 Hit Ratio by Quoting System Table All 8 Hit Ratio measures

7 Hit Ratio by Product Line Clustered Column Chart Hit_Ratio_CL, Hit_Ratio_PL

7. Binning Decision

Decision: Binning is not appropriate for this dataset.

The following rationale supports this decision:

- **RETENTION_RATIO** had 60.93% missing values — binning would produce heavily unbalanced groups that misrepresent agent performance
- **LOSS_RATIO** had 24.02% missing values — bins would not be representative of the true distribution
- **Growth rate columns** are computed continuous attributes intended for use as independent variables in a predictive model — binning would reduce their precision and predictive value
- The dataset's primary purpose is **agent segmentation in a supervised predictive framework** — continuous variables provide more granular signal for modelling than binned categories
- The data covers a **six-state region** with limited geographic variation — geographic binning would not add analytical value

For future modelling purposes, if binning is reconsidered, it should only be applied after imputing or resolving the missing value issue in RETENTION_RATIO and LOSS_RATIO.

8. Key Findings

- **Commercial Lines vs Personal Lines:** Premium is split almost equally — CL at approximately 50.26% and PL at 49.74% of new business written, indicating a balanced product portfolio
- **Retention Ratio:** Multiple agencies achieved a perfect retention ratio of 1.0, indicating strong policy retention performance. The top 10 agencies were selected based on premium volume as a tie-breaker

- **State Performance:** The dataset covers 6 states — PA, WV, OH, MI, KY and IN. Retention ratios are consistently high across all states, suggesting strong overall performance in the region
- **Hit Ratios:** The CL-MDS quoting system recorded the highest hit ratio at 0.78, making it the most efficient quoting system. In contrast, PL-eQTte recorded the lowest at 0.02
- **Growth Trend:** Premium growth rates showed variation across the 2006–2013 period, with notable fluctuations likely reflecting broader economic conditions including the 2008 financial crisis period
- **Data Quality:** A significant proportion of pre-computed metrics in the source data contained placeholder values (99999), requiring recomputation of key metrics directly from raw transactional columns

10. Conclusion

The Power BI dashboard successfully addresses all project requirements. Data was thoroughly cleaned, key metrics were computed from source columns, and interactive visualisations were built to support agent segmentation analysis.